

The company "SustainaFund" wants to select stocks to form a portfolio for their mid-term asset allocation strategy. They manage funds and have an available budget of 100M\$ to invest in the stock market. A list of stocks is available, and they wish to select the stocks to be purchased, and the fraction of the budget allocated to each selected stock.

Each stock has an expected return for the coming investment period, belongs to a geographic region (United States or Europe), and has an ESG (Environmental, Social, and Governance) score. To be aligned with their financial sustainability goals, the selected portfolio should satisfy the following conditions:

- The investment in any particular stock should not exceed 20% of the available budget and should not be lower than 1% of the total budget.
- The total investment in any single region should not exceed 60% of the available budget (so at least 40% is invested in the other region).
- The number of different stocks to be purchased needs to be at least 30.
- The ESG score amongst the selected stocks, weighted by fraction, needs to be at least 70.

Input Data

The dataset is built from real market data (STOXX Europe 600 and S&P 500 constituents) sourced from the LSEG platform: ten years of daily prices (2015-2025) and annual ESG scores.

Two instances are provided, each consisting of two files:

- `data_test` (~60 stocks): a small instance for debugging and testing.
- `data_full` (~1080 stocks): the full universe for the case study.

The input file 'sharesX.csv' provides, in tabular form, for each stock:

- **Stock:** The stock ticker (e.g. AAPL.OQ, ULVR.L, SAP.DE).
- **Region:** The geographic region the stock belongs to (United States or Europe).
- **Country:** The country of the stock's primary listing (provided for analysis; the diversification constraint uses Region).
- **ESG score:** The Environmental, Social, and Governance score (0-100), evaluating a company's sustainability and ethical impact.

The file 'stockpricesX.csv' provides the daily closing price of each stock over the available history (up to ten years, 2015-2025). Each row is a stock and each column is a trading day.

The expected return of each stock is not given directly: you are expected to estimate it from the historical prices as part of your analysis (for example, from the mean of the daily returns, suitably annualised). The choice of estimation method and time window is yours to justify.

Note on the data: *This is real data and contains natural imperfections to handle: some ESG scores are missing, some stocks will turn out to have negative expected returns, and prices are quoted in different currencies (EUR, USD, GBP, etc.). Part of the task is to explore and clean the data appropriately.*

Return vs Risk

The company seeks your expertise to help them select the portfolio to ensure maximum returns while minimizing the risk inherent to stock market volatility. These are two competing objectives:

- **Maximize the expected return** of the portfolio (the weighted sum of the selected stocks' expected returns).
- **Minimize the risk** of the portfolio, where risk is derived from the fluctuations in the stock prices.

Since higher returns generally come with higher risk, there is no single best portfolio but a trade-off between the two. The decision-makers would like to evaluate multiple solutions and compare outcomes for different risk profiles (e.g. risk averse, neutral, and risk prone), for example by maximizing return subject to a maximum risk level, minimizing risk, or combining both objectives. They would also like to evaluate the impact of the ESG constraint on the potential returns.

Computing returns and risk

A central part of this case study is deciding how to measure both the expected return and the risk of each stock and of the portfolio, starting from the raw price history. There is no single right answer: different choices lead to different models and different portfolios, and exploring this is part of the work. You are encouraged to try more than one approach and compare the results.

Expected return. A common starting point is to derive daily returns from the price series and annualise their mean. For a stock with prices $P(t)$, the daily return can be computed as a simple percentage change, $(P(t) - P(t-1)) / P(t-1)$, or as a log-return, $\ln(P(t) / P(t-1))$. The expected annual return is then the mean daily return times the number of trading days in a year (around 252). You also need to decide which time window to use: the full history is more stable, while a recent window is more representative of current conditions.

Risk. Some commonly used risk measures you may consider:

- **Standard deviation** of a stock's returns, or its coefficient of variation (standard deviation divided by the mean), which is comparable across stocks with different price levels.
- **Sharpe ratio**, the mean return divided by its standard deviation, capturing risk-adjusted return.
- **Portfolio variance from the covariance matrix**, which measures the variance of the whole portfolio using the covariance between every pair of stocks. Unlike the per-stock measures above, this captures how stocks move together: combining stocks that do not move in sync reduces overall risk (diversification). This is the basis of the classic Markowitz model.

Per-stock measures lead to simpler (linear) models, while the covariance-based measure leads to a richer (quadratic) one. Trying both and comparing is encouraged.

Tasks

Preparation phase (August 24–28, 2026, online):

1. Understand the business problem and think about your strategy:
 - a. Read and investigate the problem and related portfolio optimization solution methods.
 - b. Define the algorithmic strategy, objectives and success criteria for your project.
2. Data preparation and cleaning:
 - a. Exploratory data analysis, handling missing data, outliers, different currencies, etc.
 - b. Decide on a risk measure and how to estimate it from the price history.
3. If using Python, get familiar with the Python interface of the FICO Xpress solver:
 - a. See `fico-xpress/python-notebooks` for modeling examples using Jupyter notebooks.
 - i. You can use Github Codespaces to execute the files, as explained in the README.
 - b. Check the Xpress Python API reference manual, the Xpress Python notebook examples, and the video series on the FICO Xpress Python API.
 - c. If you are already familiar with modelling languages such as AMPL, GAMS or Xpress Mosel, you can use any of these instead of the Python API.
4. Online introduction to the case study by a member of FICO, alongside the accompanying lecture on portfolio optimization.
5. Meet with members of FICO on August 26 at 2pm CET.

On-site phase (August 31 – September 4, 2026, in Berlin):

1. Model building and experimentation:
 - a. Write the (mathematical) optimization model formulation(s) and implement them using e.g. the Python API of the FICO Xpress Solver.
 - b. Computational experiments, testing, tune parameters, run scenarios.
 - c. (The small data_test instance is provided for debugging and testing before scaling to data_full.)
2. Extending the model (optional, encouraged):
 - a. You are welcome to make the model more realistic by bringing in additional data of your own. For example, you could fetch sector or industry classifications for the stocks and add a sector diversification constraint, include transaction costs, currency effects, or any other constraints and objectives you find relevant.
 - b. Any external data you use should be documented and, ideally, reproducible.
 - c. Some data sources you may use (availability and terms of use may change, please check each source): Yahoo Finance via the yfinance Python package (free; prices, sector/industry, fundamentals); Alpha Vantage (free API key; prices, fundamentals, sector); Wikipedia constituent lists such as the S&P 500 (company-to-GICS-sector mappings); Bloomberg terminal, if you have access, e.g. through your university (comprehensive market, fundamental, and ESG data).
3. Dashboard with visualization elements that shows:
 - a. Input data insights, and results using charts, tables, and/or other visualization elements.
 - b. Simulate that you present your findings to the upper management team of the company.
 - c. (Ideally the dashboard is interactive to allow running the model for alternative user-defined scenarios or input parameters.)
 - d. You can use FICO Xpress Insight or open-source software (e.g. Streamlit, Jupyter Notebooks...).
 - e. See online material for Insight (only works with python (.py) source files, not compatible with Jupyter notebooks).
4. Online consultations with members of the FICO Xpress dev. team on September 1 at 4pm CET and September 3 at 10am CET.

Model freeze deadline

Your optimization model and the chosen portfolio must be finalized by **Thursday 3 September, 5pm CET**. After this deadline, **no further changes to the model or the solution (the selected portfolio) are allowed**. You may still refine your presentation, slides, and user interface / dashboard, but the model and the resulting portfolio are frozen.

Literature sources:

- Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77-91.
- Palomar, D. P. (n.d.). Portfolio Optimization: Theory and Application. Cambridge University Press. <https://bookdown.org/palomar/portfoliooptimizationbook/>
- Milhomem, D., & Dantas, M. (2020). Analysis of new approaches used in portfolio optimization: A systematic literature review. <https://www.redalyc.org/journal/3967/396762077018/html/>
- Kapadia, J., et al. (2021). Operations Research Applied to the Financial Markets. IJISRT.

Installation guidelines:

To use the FICO Xpress Python interface, you need a Python version between 3.10-3.14, and you can install the package using pip:

pip install xpress

or conda: conda install -c fico-xpress xpress

Licensing: FICO facilitates a full Xpress license for the participants working on this case study during the period of the summer school. You will receive a license file named 'xpauth.xpr' via e-mail, which you should:

- (if working on the cloud) upload it into the root directory of your private codespace or project.
- Set an environment variable named XPAUTH_PATH to (the path to) the license file 'xpauth.xpr'. In python:

```
import os
```

```
os.environ['XPAUTH_PATH'] = 'xpauth.xpr'
```

The provided license must only be used for the purposes of the HTW summer school. Please join the FICO Academic Partner Program to access full licenses of Xpress free of charge.

Disclaimer: *This case study is an educational exercise. The data and any results are provided solely for learning purposes and do not constitute financial, investment, or trading advice. Nothing here is a recommendation to buy, sell, or hold any security. Past performance does not guarantee future results.*